CASE STUDY 14 — CUSTOM BUSINESS LOGIC ARCHITECTURE

Introduction to Deluge Scripting: Writing Custom Functions

LANGUAGE	FOCUS	DATA TYPES	EXECUTION
Deluge (Zoho DSL)	Custom Functions & Logic	Map, List, Int, String	On-Demand & Workflows

1. EXECUTIVE OVERVIEW

Deluge (Data Enriched Language for Universal Grid Environment) is an online scripting language integrated across Zoho services. It allows developers to add complex business logic, perform arithmetic calculations, manipulate data structures, and automate workflows without managing external servers or deployment pipelines.

Low-Code
TYPE-SAFE SCRIPTING

Built-in
MAP & LIST DATA STRUCTURES

Reusable
STANDALONE FUNCTIONS

2. DELUGE SCRIPTING EXECUTION FLOW

Deluge custom functions take input parameters, execute procedural control logic, manipulate key-value maps or collections, and return formatted outputs or update database records.



3. CORE DELUGE DATA TYPES & SYNTAX STRUCTURES

Deluge supports essential data primitives and structured collections used to store, parse, and transform operational data dynamically.

Data Type	Declaration Example	Primary Use Case	Common Built-in Methods
String	`name = "Zoho Creator";`	Text values and labels	`.toLower()`, `.contains()`, `.subString()`
Int / Decimal	`amount = 1250.75;`	Numeric calculations	`.round()`, `.abs()`, `.toLong()`
List	`items = List();`	Ordered array of elements	`.add()`, `.get()`, `.size()`
Map	`userMap = Map();`	Key-value pairs / JSON	`.put()`, `.get()`, `.keys()`

4. WRITING A CUSTOM DISCOUNT CALCULATION FUNCTION

> **Function Scope:** Takes customer tier and order amount, calculates tiered discounts using conditional logic, and returns a structured response Map with final prices.

Deluge Function Implementation

```
// Custom Function: Calculate_Order_Discount Map calculateOrderDiscount(Decimal orderAmount, String customerTier) { discountRate = 0.0; // Tiered Discount Evaluation if (customerTier == "VIP") { discountRate = if(orderAmount > 1000, 0.20, 0.15); } else if (customerTier == "Regular") { discountRate = if(orderAmount > 1000, 0.10, 0.05); } discountVal = orderAmount * discountRate; finalAmount = orderAmount - discountVal; // Build Return Map responseMap = Map(); responseMap.put("OriginalAmount", orderAmount); responseMap.put("DiscountApplied", discountVal); responseMap.put("FinalAmount", finalAmount.round(2)); return responseMap; }
```

5. SUMMARY & BEST PRACTICES

Developing reusable custom functions in Deluge promotes clean code maintainability, reduces repetition across form actions, and simplifies integration with external REST APIs. Best practices include encapsulating repetitive tasks into functions, utilizing type casting functions like `.toString()` and `.toDecimal()`, and validating Map keys before retrieval.